

Tide retrospective: the finite-order smooth recovery (smooth-germ programme, stages C4–C5)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The closing tide of the smooth-germ programme, and with it the germbij nondegenerate arc: the bridge from the polynomial recovery machinery to smooth losses, which is where Theorem 3.1 actually lives. The consult’s staging held: with the local Taylor comparison in place, C4 was composition plus one genuinely new ingredient — which turned out to be a Mathlib lookup (`taylor_isLittle0`, the Peano remainder).

2 The thing formalised

The headline¹:

Two admissible C^{R+2} losses whose normalized moments at x^{2+r} differ by $o(q^{2+2r})$ for every $r \leq R$ have the same Taylor data through degree $R+2$: the quadratic coefficient $\lambda/2$ and every higher coefficient.

Supporting cast, all in `Laplace/OneD/SmoothRecovery.lean`: the Peano remainder in epsilon-radius form; the derived minimum facts (the derivative vanishes at an admissible minimum, and $\rho \leq \lambda/2$, so nondegeneracy costs no hypothesis); the Taylor polynomial rewritten in jet shape; the stabilizer extension and its identity; the stabilized jet’s joint admissibility-and-profile certificate; the loss-versus-stabilized-jet epsilon bound (Peano plus stabilizer absorption at radius $\varepsilon/(2(d+1))$); and the truncated strong induction.

3 The one structural discovery

The recovery theorem of the weighted-jet programme quantifies over *every* rung of its coefficient vector. The stabilized jet is deliberately longer than the data reaches — $M - 2 > R$ rungs, the stabilizer living above the Taylor degree precisely so it cannot disturb the recovered coefficients — so the full-length theorem is inapplicable, and honestly so: the data says nothing about the stabilizer coefficients. The fix is `jet_recovery_stable_partial`, the same strong induction truncated at rung R . This is the kind of quantifier mismatch that only surfaces at composition time, and it is exactly the “full-jet quantifier management” the original scoping consult listed among C’s obligations.

¹`smooth_jet_recovery`.

4 Proof notes

The per-rung assembly follows the recorded plan: the jet-moment difference at observable x^{2+r} , scaled by q^{-r} , decomposes through the scaling bridge into three pieces — two loss-versus-polynomial comparisons (each a C3 limit times q^{D-2-r} , vanishing or bounded), and the data hypothesis verbatim in the middle. The pow-splitting rewrites ($q^{2+r+D-2} = q^{2+2r} \cdot q^{D-2-r}$, proven by `omega` on the exponents) keep natural-number subtraction away from `ring`. One tactical lesson worth the catalogue: the ascribed type of a large composed `Tendsto` term cannot realistically match the term-mode composite’s exact shape (associativity and negation placement); dropping the ascription and letting `congr`’ carry the function shape — with the eventual identity closed by `field_simp` and `ring` — is the robust pattern.

5 Deliberation and check

Prior-deliberation path: C4–C5 were specified in the scoping consult and the composition plan was recorded in the tide log before writing. No new closed form entered; the analytic content was checked at C3 (statement-level) and across the weighted-jet programme’s executed checks.

6 What this closes, and what remains

With this tide the germbij nondegenerate programme is formalised end to end: identifiability (Theorem 7.3, 1D and $\mathbb{R}^d$), the anharmonic and weighted-jet recovery ladders, Proposition 4.1 with its witness, the polynomial comparison core with unequal bases, and now the smooth finite-order recovery. The note’s remaining commissions are the multivariate Theorem 3.1 (the Susceptibility Primer’s missing Stage 3, needing Wick machinery on $\mathbb{R}^d$) and the full-jet quantifier packaging (C6), both needing fresh scoping. The natural next act after the merge is the note-side markers on Theorem 3.1 for `smooth_jet_recovery`.